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Article · Energy efficiency

Energy audit before solar: Why South African businesses should cut waste first.

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In short

- Electricity prices rose again in 2026: 8.76% for Eskom's direct customers from April, and 9.01% on what Eskom charges municipalities from July.
- The US EPA estimates that around 30% of the energy used in commercial buildings is wasted.
- Solar sized on a wasteful load costs more and pays back later. Measure and remove the waste first.
- Every kWh saved avoids roughly 0.94 kg of CO_2 on South Africa's grid.
- The Section 12L energy efficiency tax incentive now runs to 31 December 2030, but it needs a baseline measured before you make changes.

Rooftop solar has become the default answer to rising electricity bills in South Africa. It is often a good answer. But most commercial and industrial systems are sized on the bill the building pays today, and that bill includes everything the building wastes: Air conditioning running for empty floors, compressors leaking air all weekend, lights burning in warehouses at 2 a.m., and demand peaks set by equipment starting at the same moment.

Size solar on that load and you pay to generate power that is then thrown away. This guide explains why an energy audit should come first, what has changed in 2026, and what a good audit actually measures.

Electricity is one of your fastest-rising costs

In March 2026 the National Energy Regulator of South Africa (NERSA) approved higher increases for 2026/27 than first planned, after correcting a R54.7 billion error in Eskom's regulatory asset base. The result is an 8.76% increase for customers supplied directly by Eskom from 1 April 2026, and a 9.01% increase in what Eskom charges municipalities from 1 July 2026. Municipalities pass that through in their own tariffs, which they set individually, so your actual increase depends on where you are and which tariff you are on.

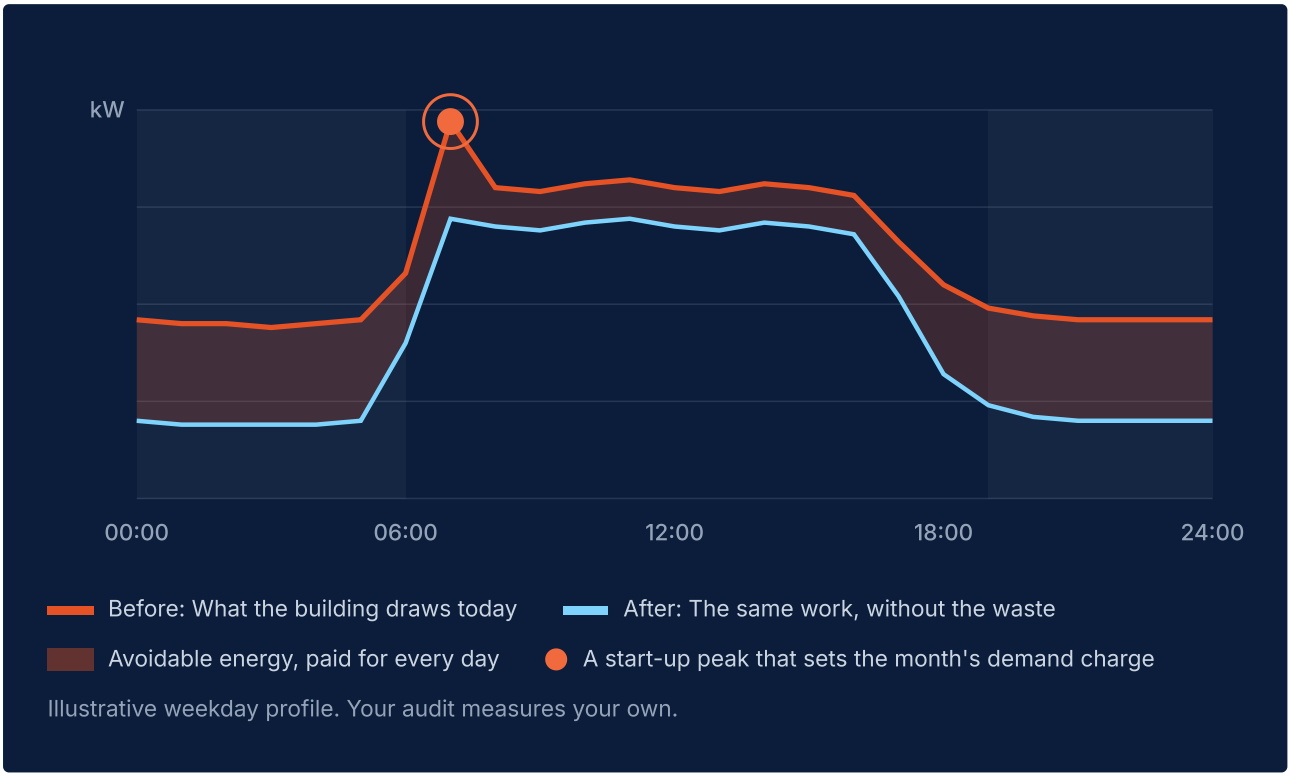
For a business, that matters twice. The energy charge per kWh rises, and so do the demand and network charges many commercial tariffs add on top. Waste that was tolerable at last year's prices is a bigger line item this year, and it compounds.

Most buildings waste more than anyone has measured

ENERGY STAR, the US Environmental Protection Agency's programme, estimates that around 30% of the energy used in commercial buildings is wasted. South African buildings are no

different in kind: The same air conditioning, lighting, refrigeration, compressed air and motor loads, often controlled by timers and setpoints nobody has checked since commissioning.

The problem is visibility. Electricity arrives as one number at the end of the month. Nobody can tell you which plant, which board or which hour of the night drove it. The chart below shows the pattern we look for first.



The shaded band is energy the building pays for while it is empty. The spike at 07:00 is a start-up peak: On a tariff with a monthly kVA demand charge, that single moment can set what you pay for capacity for the entire month.

Where the waste usually hides

- **After-hours base load:** HVAC, extraction, lighting, compressors and process equipment left running at night and over weekends.
- **Maximum demand peaks:** Large loads starting together, which solar cannot fix if they happen before sunrise.
- **Poor power factor:** Reactive energy charges and inflated kVA demand. Correcting a site from 0.91 to 0.99 is often one of the fastest paybacks available.
- **HVAC and refrigeration:** Wrong setpoints, simultaneous heating and cooling, and poorly sequenced chillers.
- **Compressed air:** Leaks and over-pressure, frequently the most expensive utility on an industrial site.
- **Motors, pumps and fans:** Oversized and throttled instead of speed-controlled.
- **The electrical network:** Overloaded cables, lightly loaded transformers, harmonics and phase imbalance that turn paid-for energy into heat.

Why efficiency should come before solar

Solar reduces the energy you buy while the sun shines. It does nothing for the load you carry at night unless you add batteries, and batteries are the most expensive part of a hybrid system. Cut the waste first and four things happen:

1. **The PV system can be smaller** for the same share of your daytime load, or the same system covers more of it.
2. **The battery can be much smaller,** because the overnight load it has to carry is lower.
3. **Demand charges fall,** which solar alone often fails to touch when the peak happens at dawn.
4. **The payback starts sooner,** because efficiency measures such as controls, scheduling and power factor correction usually cost less per kWh saved than new generation.

Globally, efficiency is the area most countries are behind on. The International Energy Agency reported in November 2025 that global energy efficiency is improving by about 1.8% in 2025, up from around 1% in 2024, but still well short of the 4% a year that nearly 200 governments agreed to aim for by 2030. Buildings and industry are where that gap is closed, one site at a time.

The climate case: Every kWh you don't use

South Africa's official grid emission factor for domestic generation was 0.942 tonnes of CO_2 equivalent per MWh in 2023, or roughly 0.94 kg for every kWh. For most commercial buildings, purchased electricity is the largest part of the carbon footprint, so wasted electricity is wasted carbon.

The pressure to measure it is growing. The headline carbon tax rate rose to R308 per tonne of CO_2 equivalent from 1 January 2026. It is levied on direct emitters rather than on electricity users, but it shapes what large customers, lenders and export markets now ask their suppliers to report. A measured energy baseline is the starting point for that reporting.

Two rules that make measurement worth money

Section 12L: 95 cents per kWh saved, until 2030

The Section 12L energy efficiency tax incentive lets a business deduct 95 cents for every verified kWh saved from its taxable income. In August 2025 the government announced that it has been extended by five years, to 31 December 2030. The catch is verification: Savings must be measured and verified by a SANAS-accredited measurement and verification body, against a baseline set before the efficiency project is implemented. If you change the plant first and measure later, you may lose the claim. Talk to your tax adviser and arrange the baseline before any work starts.

Energy Performance Certificates

Under the national EPC regulations, privately owned buildings with a net floor area above 2,000 m², and public buildings above 1,000 m², in the office, place of instruction and public assembly occupancy classes had to display a valid Energy Performance Certificate by 7 December 2025. The rating follows SANS 1544, and non-compliance can attract fines of up to R5 million. An energy audit does not replace the certificate, but the measurements behind it show you what to fix before the next rating.

What a proper energy audit measures

An energy audit is only as good as its data. A walk-through with a checklist will find the obvious. Finding where the money actually goes takes measurement over time:

1. **Bills and tariff review:** Twelve to twenty-four months of accounts, the tariff structure, time-of-use periods, notified maximum demand and reactive energy charges.
2. **Site walk-down:** Tracing the main loads with the people who run the building.
3. **Logging:** Power quality analysers on the main incomer and key distribution boards for one to two weeks, so the profile covers real weekdays, nights and weekends.
4. **Load flow and loss analysis:** Where the energy goes, board by board and hour by hour, and where the network itself loses it.
5. **Feasibility study:** Each measure costed and ranked by savings, payback and carbon, including its effect on any planned solar or battery system.
6. **Verification plan:** A method agreed up front so savings are proven against the baseline.

A worked example

Consider a commercial building that the audit shows is carrying 40 kW of avoidable load outside working hours: Air conditioning, lighting and ventilation that serve nobody. With 14 unoccupied hours on each of 22 weekdays and 24 hours on each of 8 weekend days, that is about 500 hours a month.

Illustrative figures, not a quote

Avoidable energy	40 kW × 500 h ≈ 20,000 kWh a month, 240,000 kWh a year
Cost at an assumed R2.20 per kWh	≈ R44,000 a month, R528,000 a year
Carbon at 0.942 kg per kWh	≈ 226 tonnes of CO_2 a year
Possible Section 12L deduction	240,000 kWh × R0.95 = R228,000 of deductions, about R61,500 less tax at a 27% company rate, if verified
Battery you no longer need	40 kW × 14 h ≈ 560 kWh of overnight storage per weekday

Your tariff, hours and loads will differ, which is exactly why the audit measures them. The point is the order of magnitude: The waste is often worth more than people expect, and removing it

changes what the right solar system looks like.

Before you sign a solar quote, ask

- What is our base load at night and over weekends, and how much of it is necessary?
- When does our monthly maximum demand happen, and will solar reduce it?
- What is our power factor, and are we paying reactive energy charges?
- Was the system sized on measured load profiles, or on the bill?
- Has a baseline been measured, in case we want to claim Section 12L?
- Who will commission the system and sign the SSEG report for the utility?

Find out where your electricity goes

Bynergex runs Energy Loss Audits for commercial and industrial buildings in Cape Town and across South Africa, and commissions and signs off solar systems once they are installed.

Frequently asked questions

Should I get an energy audit before installing solar?

Yes. A solar system sized on a wasteful load is bigger and slower to pay back. Measuring and removing after-hours load, demand peaks and power factor penalties first means a smaller PV system and battery for the same result.

How much does energy waste cost a commercial building?

It depends on the site, but the US EPA estimates around 30% of the energy used in commercial buildings is wasted. In our illustrative example, 40 kW of avoidable after-hours load costs about R528,000 a year at R2.20 per kWh.

Is the Section 12L energy efficiency tax incentive still available?

Yes. The government announced in August 2025 that Section 12L has been extended to 31 December 2030. It allows a deduction of 95 cents per verified kWh saved, measured by a SANAS-accredited measurement and verification body against a baseline set before the project.

Does an energy audit give me an Energy Performance Certificate?

No. An EPC is issued under SANS 1544 by an accredited professional. An energy audit measures where energy is used and wasted, which shows what to improve before the building is rated.

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